



MOVIE RATING DB

Business Intelligence and Big Data Analytics





DATASET

MovieLens 1M Dataset

MovieLens 1M movie ratings. Stable benchmark dataset. 1 million ratings from 6000 users on 4000 movies. Released 2/2003.

- [README.txt](#)
- [ml-1m.zip](#) (size: 6 MB, [checksum](#))

Permalink: <https://grouplens.org/datasets/movielens/1m/>

01

MovieLens 1M Dataset of [GroupLens](#) (Minnesota University)

02

Includes >1 million ratings from 6,000 users for 4,000 movies.

03

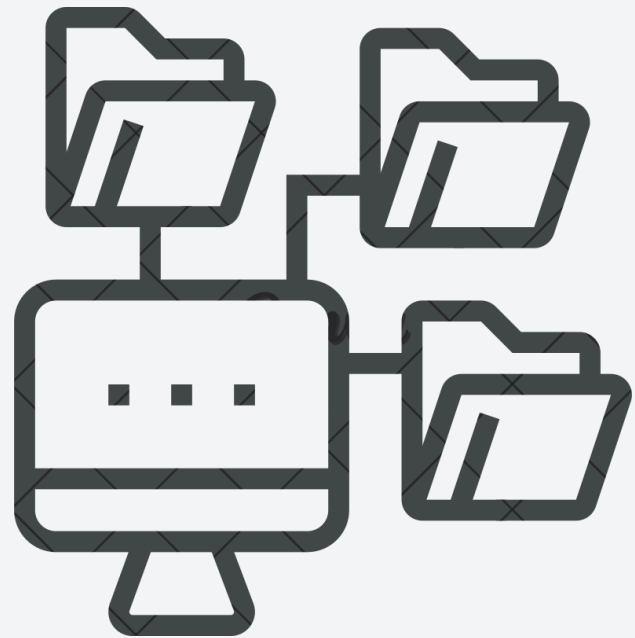
Consists of 3 files.

- > **ratings.dat**
- > **movies.dat**
- > **users.dat**

04

Source: <https://grouplens.org/datasets/movielens/1m/>

DATASET: FILES



Ratings file (1000209 instances)

› **UserID::MovieID::Rating::Timestamp**

Movies file(3952 instances)

› **MovieID::Title::Genres**

Users file(6040 instances)

› **UserID::Gender::Age::Occupation::Zip-code**



DATASET: RATINGS FILE

- Consists of:
- MovieID*** > *type: int*
values: movie's code
 - MovieID*** > *type: int*
values: movie's code
 - Rating*** > *type: int*
values: 1-5
 - Timestamp*** > *type: int*
values: time record of rating



DATASET: MOVIES FILE

Consists of: **UserID**

> *type: int*
values: user's code

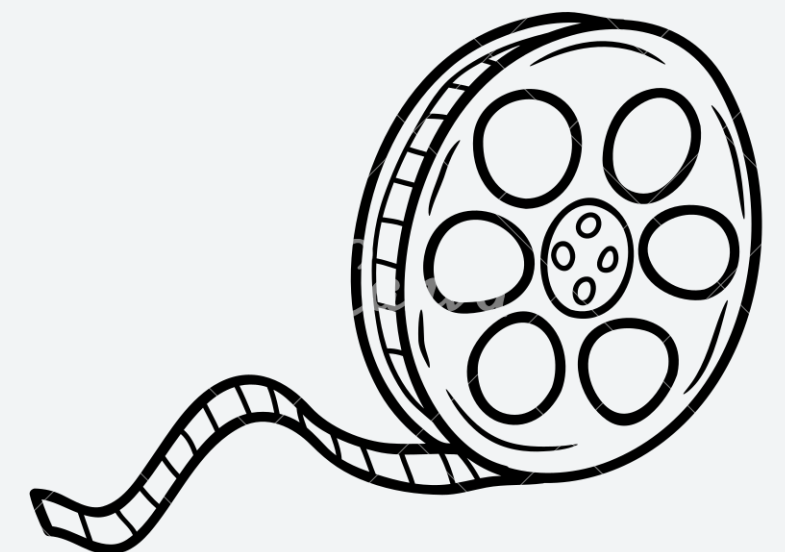
Title

> *type: string*
*values: movie's title (with
release year in brackets)*

Genres*

> *type: string*
*values: movie's genres
seperated by "|"*

* **Genres:** Action, Adventure, Animation, Children's, Comedy, Crime, Documentary, Drama, Fantasy, Film-Noir, Horror, Musical, Mystery, Romance, Sci-Fi, Thriller, War, Western



DATASET:USERS FILE

Consists of: **UserID**

> *type: int*
values: user's code

Gender

> *type: string*
values: m (Male) or f (Female)

Age**

> *type: int*
values: id, refers to age

Occupation*

> *type: int*
values: id, refers to occupation

Zipcode

> *type: int*
values: user's zip-code

**** Age Ranges:**

1: Under 18,
18: 18-24,
25: 25-34,
35: 35-44,
45: 45-49,
50: 50-55,
56: 56+

*** Occupations:** other, academic/educator, artist, clerical/admin, college/grad student, customer service, doctor/health care, executive/managerial, farmer, homemaker, K-12 student, lawyer, programmer, retired, sales/marketing, scientist, self-employed, technician/engineer, tradesman/craftsman, unemployed, writer



Software we used for the project

All the software/tools listed below were used for the creation and visualization of our project



Jupyter Notebook



**SQL Server
Management
Studio**



**Microsoft Visual
Studio**



Tableau



Python

EXTRACT, TRANSFORM, LOAD (ETL)

Conversion of files from .dat to .csv using Python with Jupyter Notebook.

```
import csv

def convert_dat_to_csv(dat_file_path, csv_file_path):
    try:
        with open(dat_file_path, 'r') as dat_file:
            lines = dat_file.readlines() # Read all lines from the .dat file

        with open(csv_file_path, 'w', newline='') as csv_file:
            writer = csv.writer(csv_file)
            for line in lines:
                # Split the line using '::' as the delimiter
                writer.writerow(line.strip().split '::'))

        print(f"File converted successfully: {csv_file_path}")
    except FileNotFoundError:
        print("The specified file was not found.")
    except Exception as e:
        print(f"An error occurred: {e}")
```



```
convert_dat_to_csv(movie_dat, movie_file)
convert_dat_to_csv(rating_dat, rating_file)
convert_dat_to_csv(user_dat, user_file)
```

```
File converted successfully: C:\Users\30698\OneDrive\Desktop\ml-1m\movies.csv
File converted successfully: C:\Users\30698\OneDrive\Desktop\ml-1m\ratings.csv
File converted successfully: C:\Users\30698\OneDrive\Desktop\ml-1m\users.csv
```


EXTRACT, TRANSFORM, LOAD (ETL)

Creation of DataFrames using pandas for analysis.

```
import pandas as pd

# Read the file with column names
movie_df = pd.read_csv(movie_file, encoding='ISO-8859-1', names=['Movie ID', 'Title', 'Genre'], header=None)

# Reset index to start from 1
movie_df.index = range(1, len(movie_df) + 1)

movie_df
```

movies.csv > movie_df

Movie ID	Title	Genre
1	Toy Story (1995)	Animation Children's Comedy
2	Jumanji (1995)	Adventure Children's Fantasy
3	Grumpier Old Men (1995)	Comedy Romance
4	Waiting to Exhale (1995)	Comedy Drama
5	Father of the Bride Part II (1995)	Comedy

ratings.csv > rating_df

User ID	Movie ID	Rating	Timestamp
1	1193	5	978300760
1	661	3	978302109
1	914	3	978301968
1	3408	4	978300275
1	2355	5	978824291

users.csv > user_df

User ID	Gender	Age	Occupation	Zip-Code
1	2	1	10	48067
2	1	56	16	70072
3	1	25	15	55117
4	1	45	7	02460
5	1	25	20	55455

EXTRACT, TRANSFORM, LOAD (ETL)

Checking for null and duplicate values.

```
rating_df.isna().any()
```

User ID	False
Movie ID	False
Rating	False
Timestamp	False
dtype: bool	

```
movie_df.isna().any()
```

Movie ID	False
Title	False
Genre	False
Year	False
dtype: bool	

```
user_df.isna().any()
```

User ID	False
Gender	False
Age	False
Occupation	False
Zip-Code	False
dtype: bool	

No null values!

```
rating_df.duplicated(subset=['User ID', 'Movie ID']).any()
```

False

```
user_df.duplicated(subset=['User ID']).any()
```

False

```
movie_df.duplicated(subset=['Title', 'Year']).any()
```

False

No duplicate values!

EXTRACT, TRANSFORM, LOAD (ETL)

Separation of the release year and removal of unnecessary titles.

Seperation of the release year in a different attribute for analysis.

```
# Extract everything within brackets
bracketed_items = movie_df['Title'].str.extractall(r'\((.*?)\)')

# Count the total number of bracketed items
total_count = len(bracketed_items)

# Display the total count
print(f"Total number of bracketed items: {total_count}")

movie_df['Title'] = movie_df['Title'].str.replace(r'\((.*?)\)', '', regex=True)

Total number of bracketed items: 270
```

```
# Extract the year from the title using a regular expression
movie_df['Year'] = movie_df['Title'].str.extract(r'\((\d{4})\)', expand=False)

# Remove the year from the Title column
movie_df['Title'] = movie_df['Title'].str.replace(r'\s*(\d{4})', '', regex=True)

movie_df
```

	Movie ID	Title	Genre	Year
1	1	Toy Story	Animation Children's Comedy	1995
2	2	Jumanji	Adventure Children's Fantasy	1995
3	3	Grumpier Old Men	Comedy Romance	1995
4	4	Wild About Safety	Comedy	1995

Removal of unnecessary titles, as they are not valuable for analysis.

EXTRACT, TRANSFORM, LOAD (ETL)

Creation of associations and full date tables.

Using pandas library to convert timestamp to full date

```
movie_df['Genre'] = movie_df['Genre'].str.split('|')
df_exploded = movie_df.explode('Genre')

assosiate_df = df_exploded.merge(genre_df, on='Genre', how='left')
assosiate_df['Assosiation ID'] = assosiate_df.index + 1

assosiate_df_f = assosiate_df[['Assosiation ID', 'Movie ID', 'Genre ID']]
```

	Assosiation ID	Movie ID	Genre ID
0	1	1	3
1	2	1	4
2	3	1	5
3	4	2	2

```
data = rating_df['Timestamp']
date_df = pd.DataFrame(data)

# Convert 'Timestamp' column to a full date
date_df['Full Date'] = pd.to_datetime(date_df['Timestamp'], unit='s')

date_df['Year'] = date_df['Full Date'].dt.year
date_df['Month'] = date_df['Full Date'].dt.month
date_df['Day'] = date_df['Full Date'].dt.day
```

	Full Date	Year	Month	Day	Hour	Minute	Second
1	2000-12-31 22:12:40	2000	12	31	22	12	40
2	2000-12-31 22:25:00	2000	12	31	22	25	00

Split genres with “|” character and create a table like movieID = genreID(1), movieID=genreID(2), etc.

EXTRACT, TRANSFORM, LOAD (ETL)

Creation of the necessary tables

Genre ID	Genre
1	Action
2	Adventure
3	Animation
4	Children's
5	Comedy
6	Crime
7	Documentary
8	Drama
9	Fantasy
10	Film-Noir
11	Horror
12	Musical
13	Mystery
14	Romance
15	Sci-Fi
16	Thriller
17	War
18	Western

Movie ID	Title	Year
1	Toy Story	1995
2	Jumanji	1995
3	Grumpier Old Men	1995
4	Waiting to Exhale	1995
5	Father of the Bride Part II	1995
...	...	...

User ID	Movie ID	Rating	Timestamp
1	1193	5	978300760
1	661	3	978302109
1	914	3	978301968
1	3408	4	978300275
1	2355	5	978824291
...	...	...	...

User ID	Gender	Age	Occupation	Zip-Code
1	2	1	10	48067
2	1	56	16	70072
3	1	25	15	55117
4	1	45	7	02460
5	1	25	20	55455
...	...	...	...	...

Assosiation ID	Movie ID	Genre ID
1	1	3
2	1	4
3	1	5
4	2	2
5	2	4
...	...	...

Age ID	Age Range
1	Under 18
18	18-24
25	25-34
35	35-44
45	45-49
50	50-55
56	56+

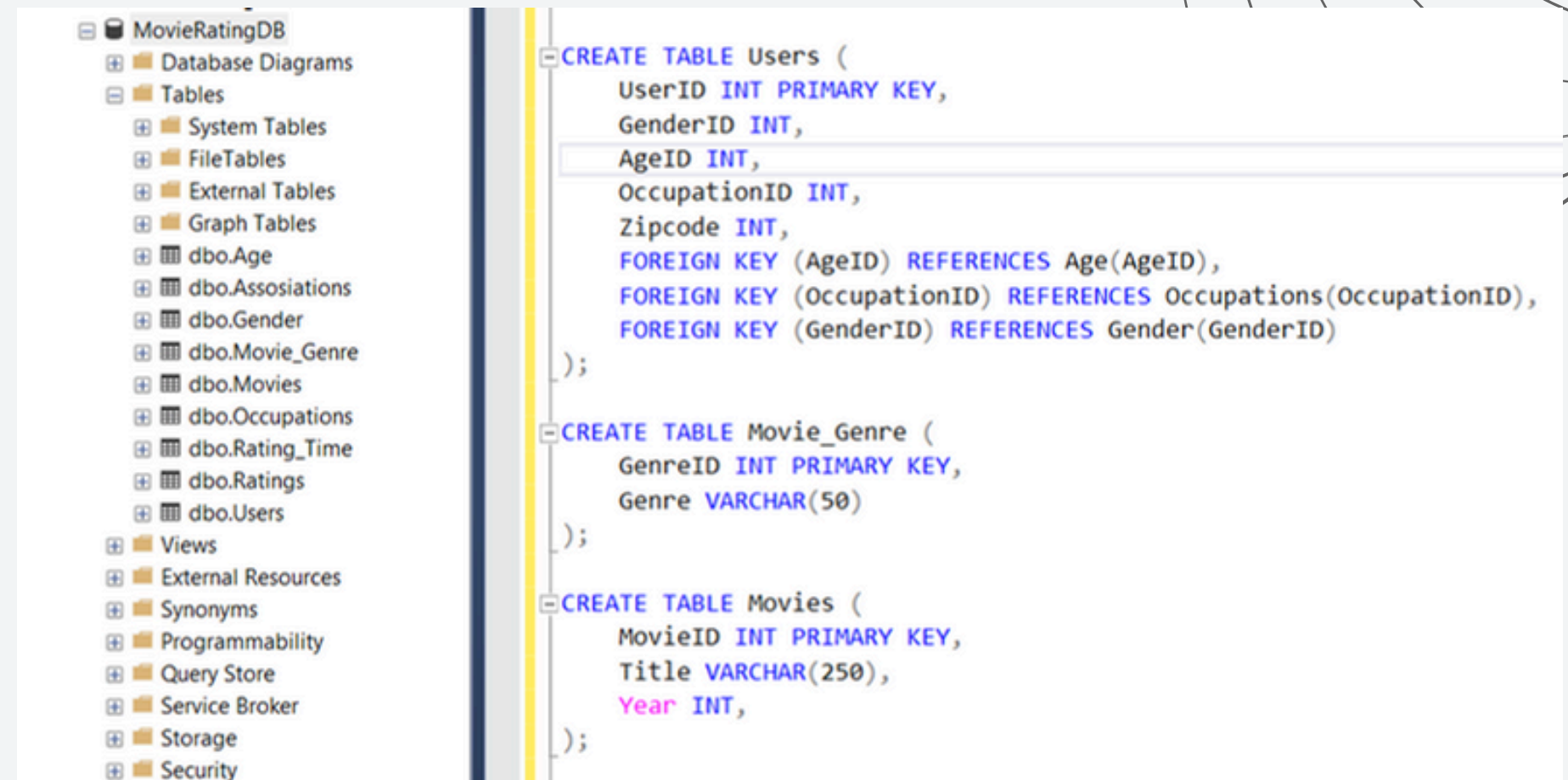
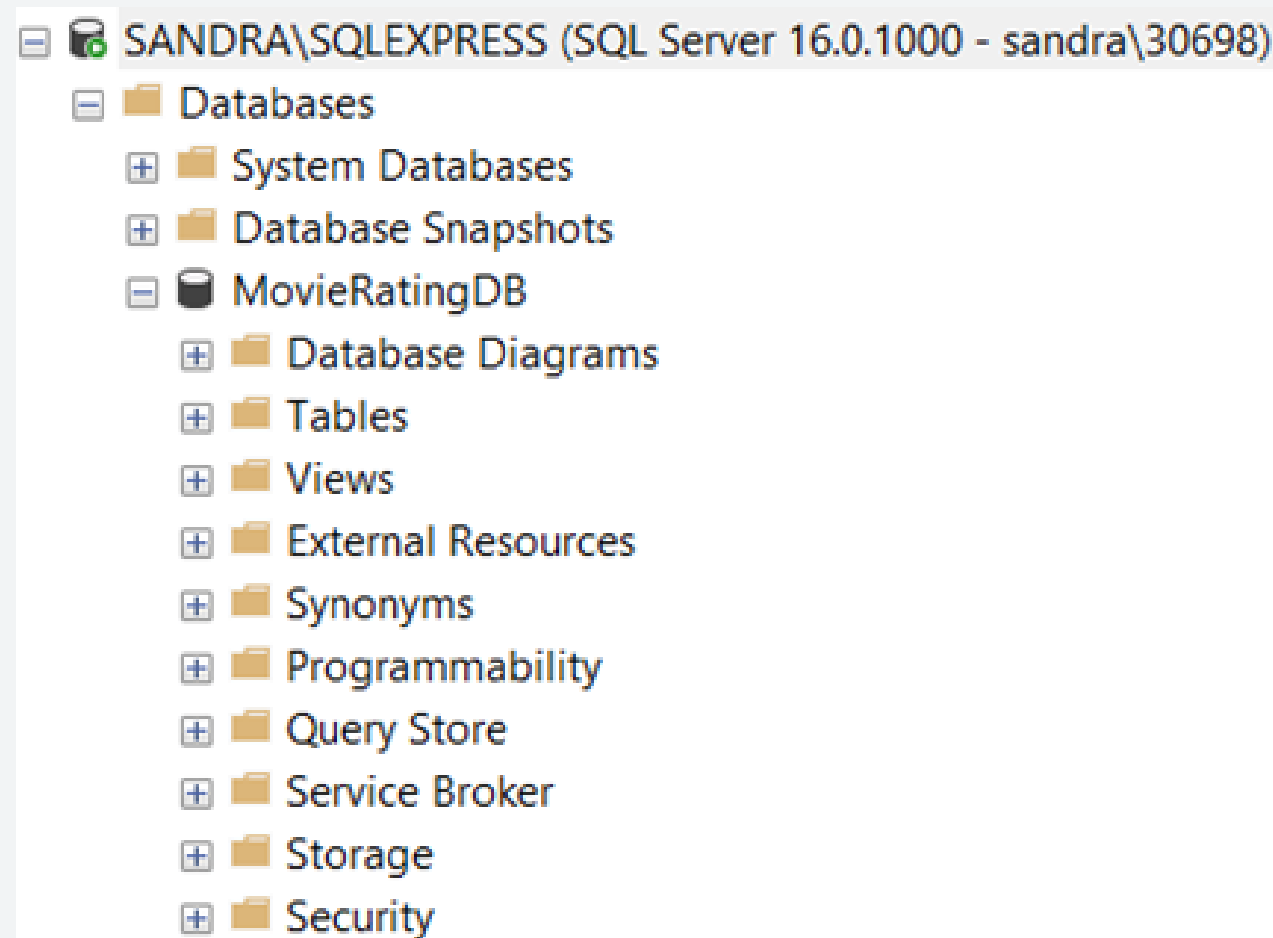
Gender ID	Gender
1	male
2	female

Full Date	Year	Month	Day	Hour	Minute	Second
2000-12-31 22:12:40	2000	12	31	22	12	40
2000-12-31 22:35:09	2000	12	31	22	35	9
2000-12-31 22:32:48	2000	12	31	22	32	48
2000-12-31 22:04:35	2000	12	31	22	4	35
2001-01-06 23:38:11	2001	1	6	23	38	11
...	...	...	...	...	...	...

Occupation ID	Occupation
0	other/not specified
1	academic/educator
2	artist
3	clerical/admin
4	college/grad student
5	customer service
6	doctor/health care
7	executive/managerial
8	farmer
9	homemaker
10	K-12 student
11	lawyer
12	programmer
13	retired
14	sales/marketing
15	scientist
16	self-employed
17	technician/engineer
18	tradesman/craftsman
19	unemployed
20	writer

INSERTING THE DATASET IN A DATA WAREHOUSE

Creating the Database using Microsoft SQL Server Management Studio



Step 1
Creating the Database

Step 2
Configuring the tables

INSERTING THE DATASET IN A DATA WAREHOUSE

Creating the Database using Microsoft SQL Server Management Studio

Step 3

Using queries to insert the data

```
USE [MovieRatingDB]

BULK
INSERT [dbo].[Movies]
FROM 'C:\Users\30698\OneDrive\Desktop\ml-1m\ratings.csv'
WITH
(
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n'
)
```



```
USE [MovieRatingDB]

BULK
INSERT [dbo].[Movies]
FROM 'C:\Users\30698\OneDrive\Desktop\ml-1m\movies.csv'
WITH
(
    FIELDTERMINATOR = '#',
    ROWTERMINATOR = '\n'
)
```

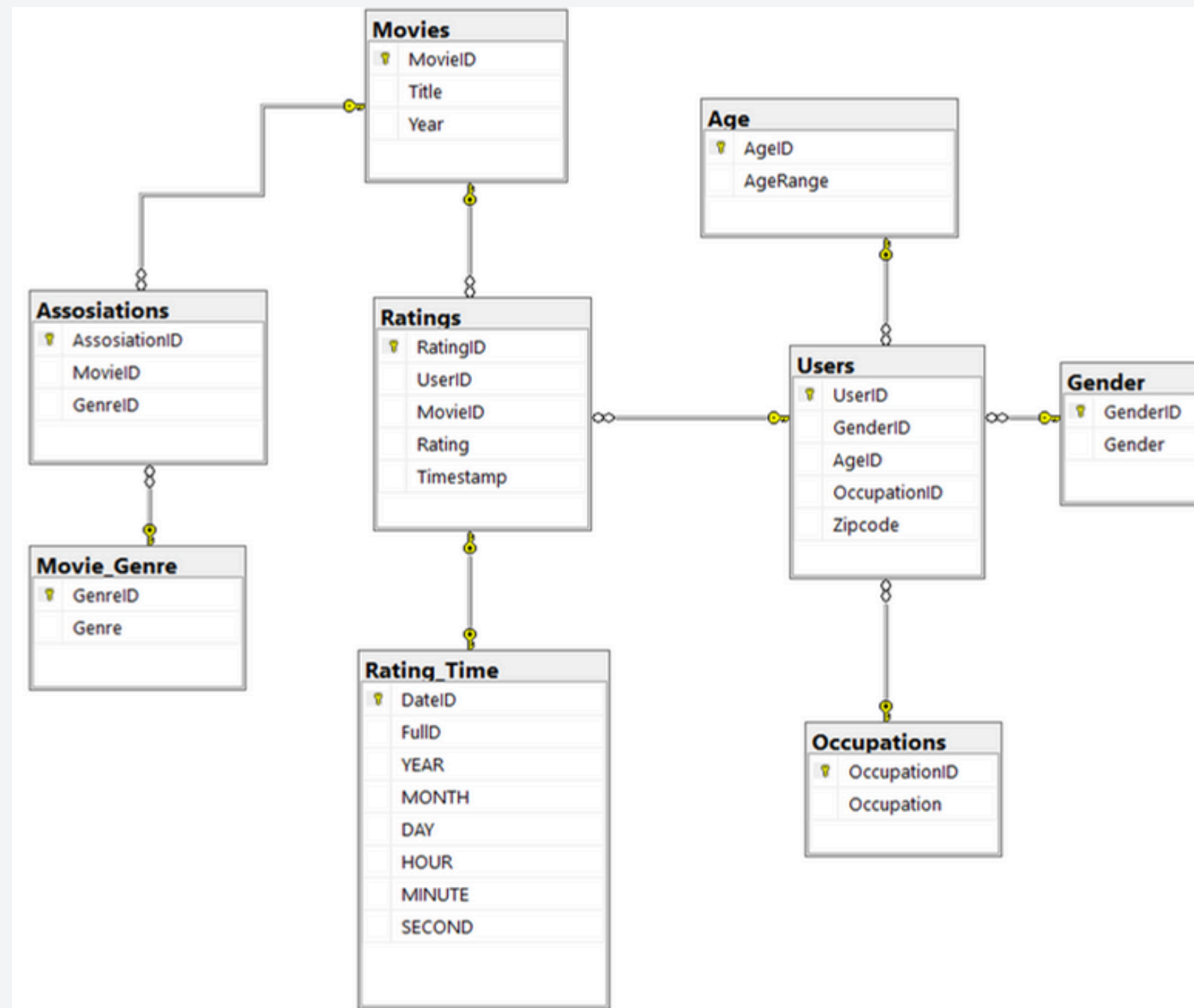
*** Using FIELDTERMINATOR='#' to ensure we avoid mistakes because of commas (,) included in the movie titles thus ensuring the data is inserted properly.**

INSERTING THE DATASET IN A DATA WAREHOUSE

Creating the Database using Microsoft SQL Server Management Studio

Step 4

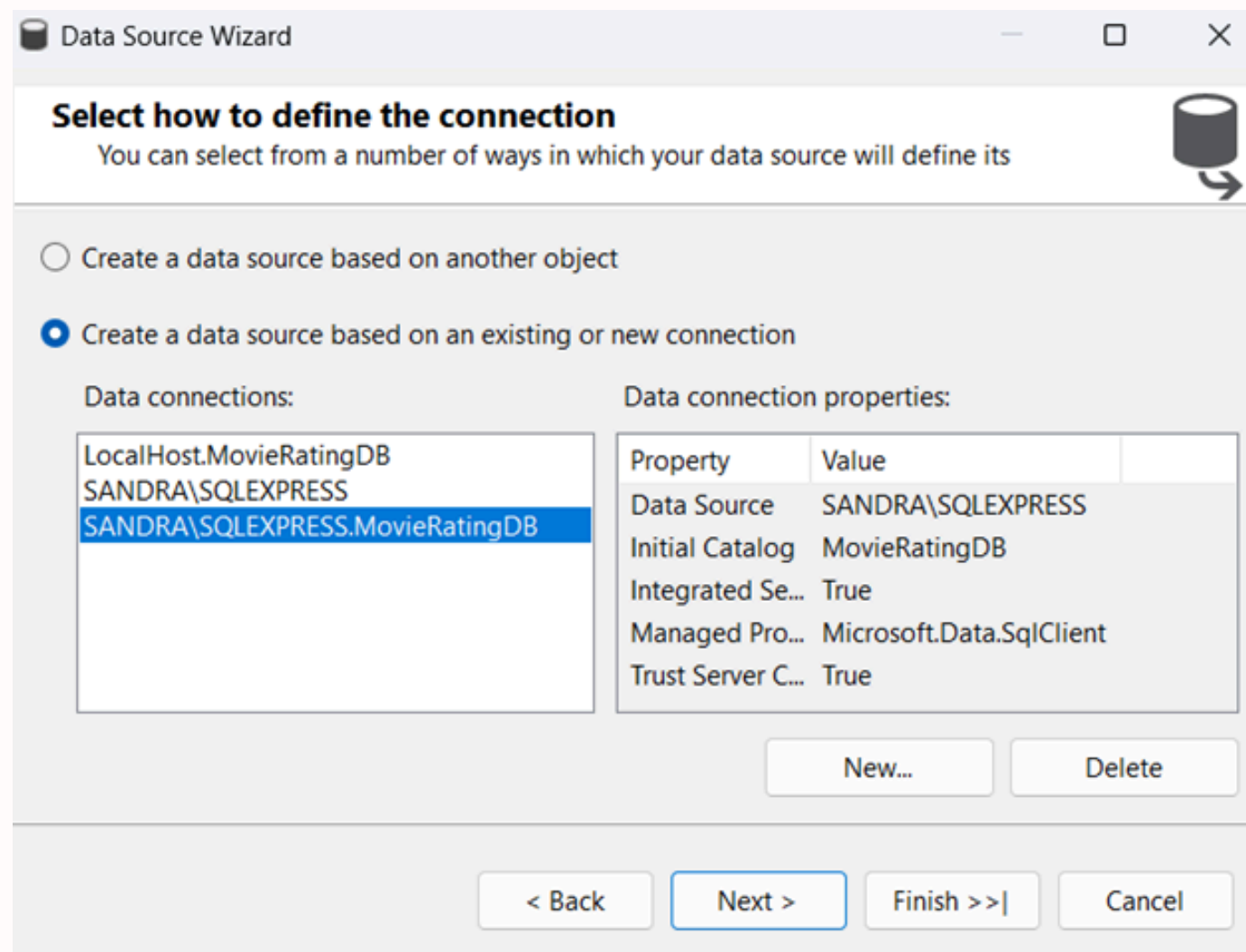
Final SQL Dabase Schema



DATA CUBE CREATION

Using Visual Studio to create an Analysis Services Multidimensional Project

Setting up Data Source Wizard



The Data Source Wizard dialog box is shown. It has a title bar with a close button. The main heading is "Select how to define the connection" with a subtitle "You can select from a number of ways in which your data source will define its". There are two radio buttons: "Create a data source based on another object" (unselected) and "Create a data source based on an existing or new connection" (selected). Below the radio buttons, there are two sections: "Data connections:" and "Data connection properties:". The "Data connections:" section contains a list box with three items: "LocalHost.MovieRatingDB", "SANDRA\SQLEXPRESS", and "SANDRA\SQLEXPRESS.MovieRatingDB" (which is selected). The "Data connection properties:" section contains a table with two columns: "Property" and "Value". The table has five rows: "Data Source" with value "SANDRA\SQLEXPRESS", "Initial Catalog" with value "MovieRatingDB", "Integrated Se..." with value "True", "Managed Pro..." with value "Microsoft.Data.SqlClient", and "Trust Server C..." with value "True". At the bottom of the dialog are four buttons: "< Back", "Next >", "Finish >>|", and "Cancel".

Select how to define the connection
You can select from a number of ways in which your data source will define its

☐ Create a data source based on another object

☒ Create a data source based on an existing or new connection

Data connections:

LocalHost.MovieRatingDB
SANDRA\SQLEXPRESS
SANDRA\SQLEXPRESS.MovieRatingDB

Data connection properties:

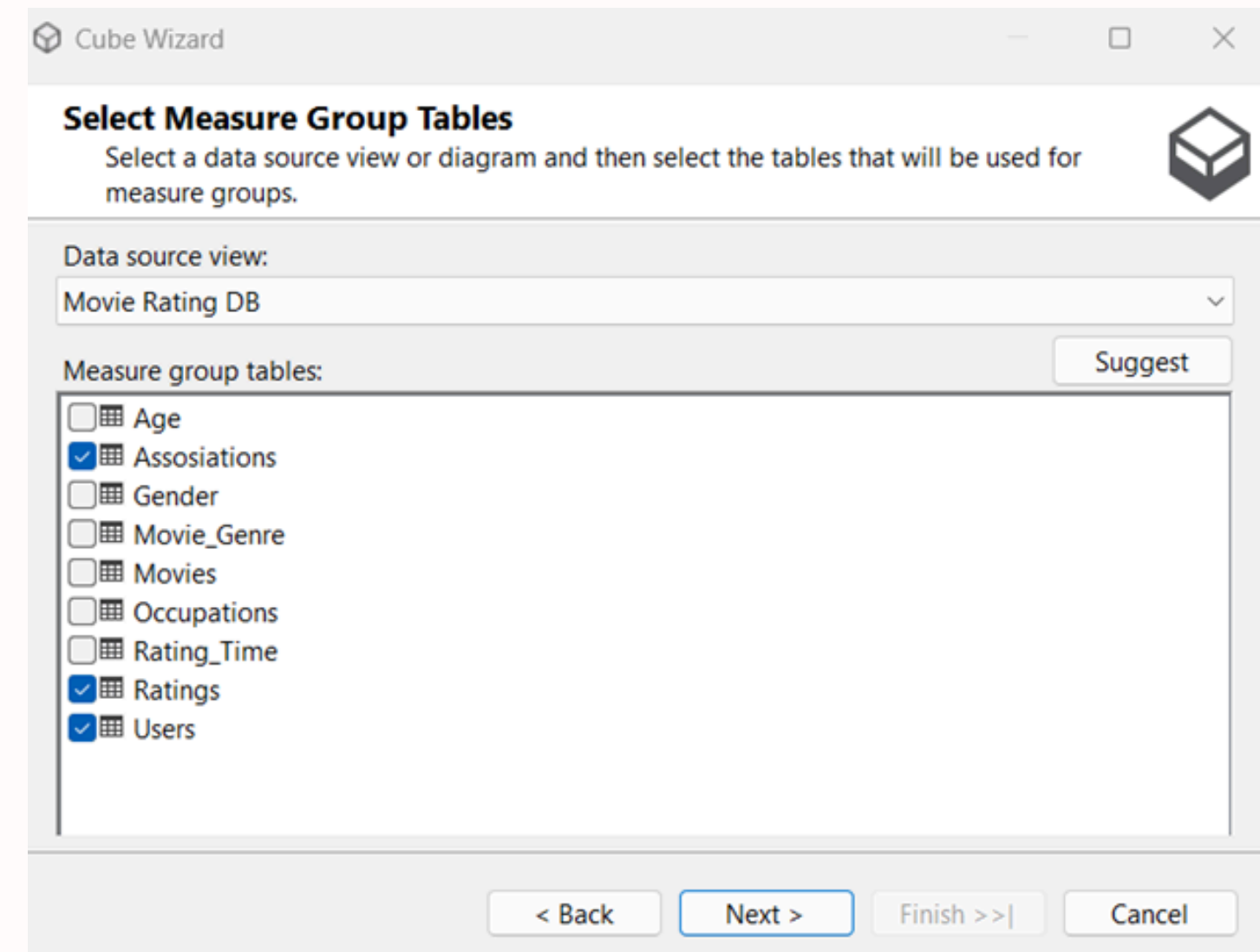
Property	Value
Data Source	SANDRA\SQLEXPRESS
Initial Catalog	MovieRatingDB
Integrated Se...	True
Managed Pro...	Microsoft.Data.SqlClient
Trust Server C...	True

New... Delete

< Back Next > Finish >>| Cancel



Selecting Tables with important information



The Cube Wizard dialog box is shown. It has a title bar with a close button. The main heading is "Select Measure Group Tables" with a subtitle "Select a data source view or diagram and then select the tables that will be used for measure groups." Below the subtitle, there is a section "Data source view:" with a dropdown menu showing "Movie Rating DB". To the right of the dropdown is a "Suggest" button. Below this is a section "Measure group tables:" with a list box containing a list of tables with checkboxes: "Age", "Associations" (checked), "Gender", "Movie_Genre", "Movies", "Occupations", "Rating_Time", "Ratings" (checked), and "Users" (checked). At the bottom of the dialog are four buttons: "< Back", "Next >", "Finish >>|", and "Cancel".

Select Measure Group Tables
Select a data source view or diagram and then select the tables that will be used for measure groups.

Data source view:
Movie Rating DB

Measure group tables:

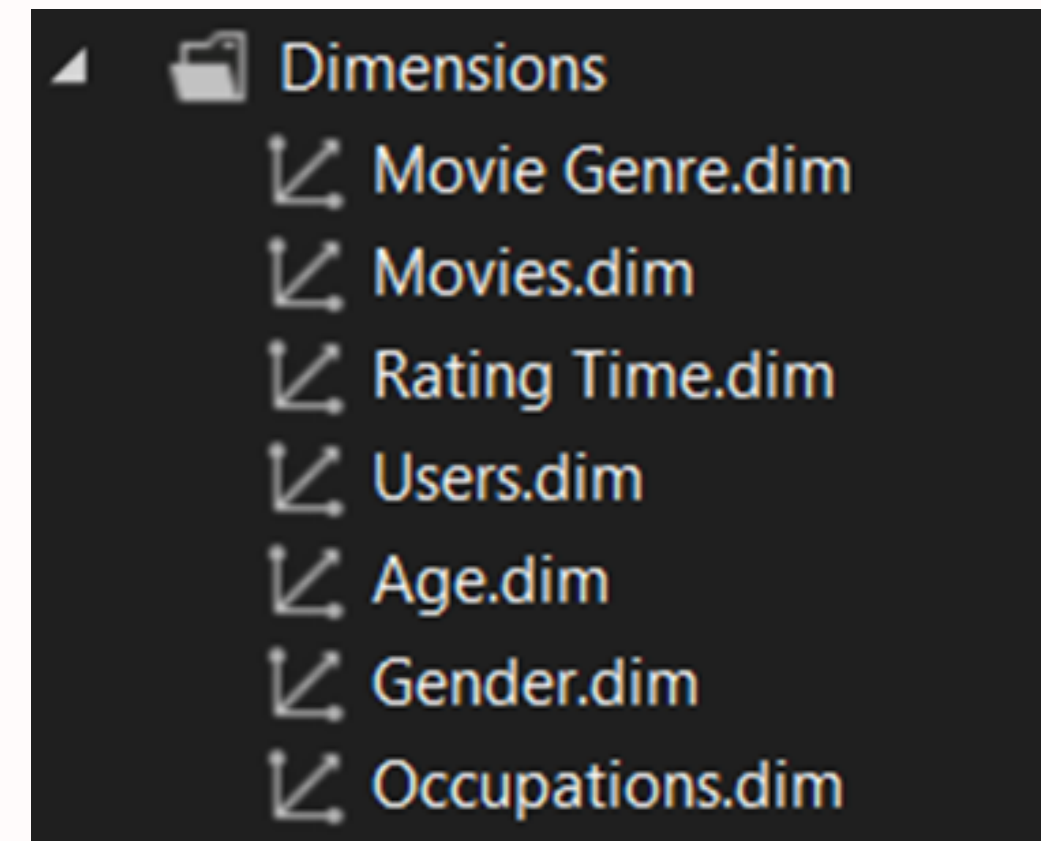
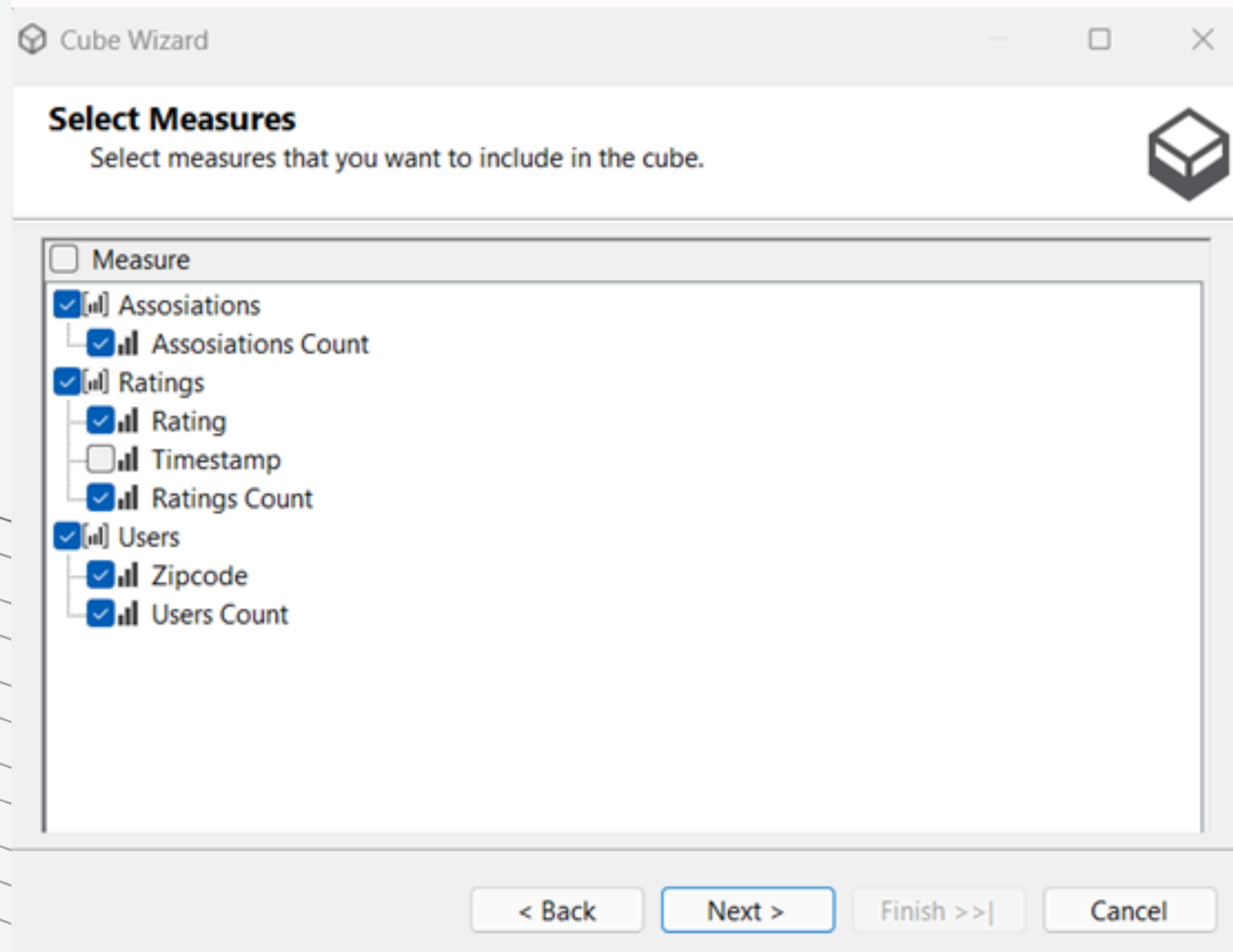
☐	Age
☒	Associations
☐	Gender
☐	Movie_Genre
☐	Movies
☐	Occupations
☐	Rating_Time
☒	Ratings
☒	Users

< Back Next > Finish >>| Cancel

DATA CUBE CREATION

Using Visual Studio to create an Analysis Services Multidimensional Project

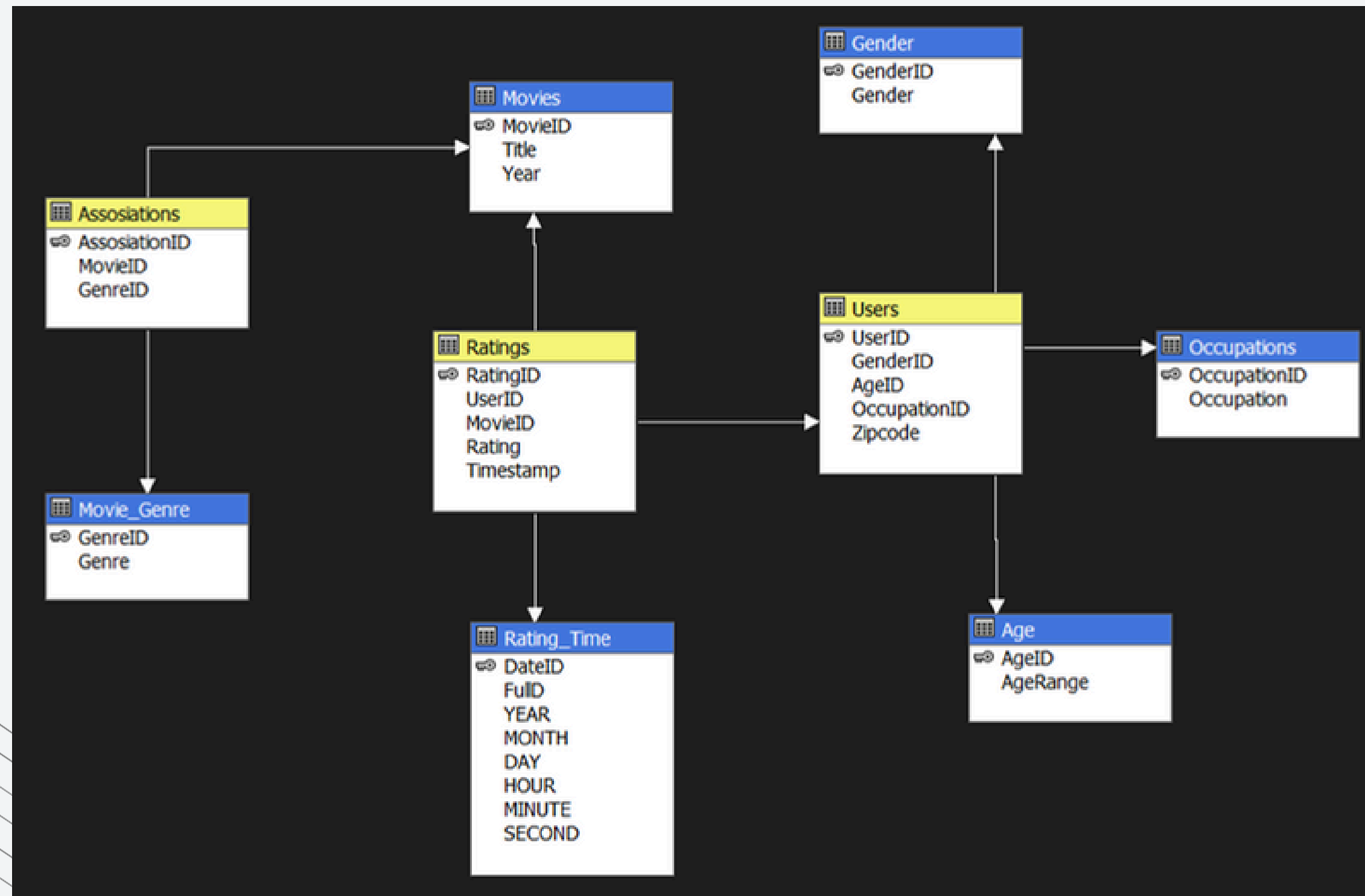
Selecting necessary measures for the data cube



DATA CUBE CREATION

Using Visual Studio to create an Analysis Services Multidimensional Project

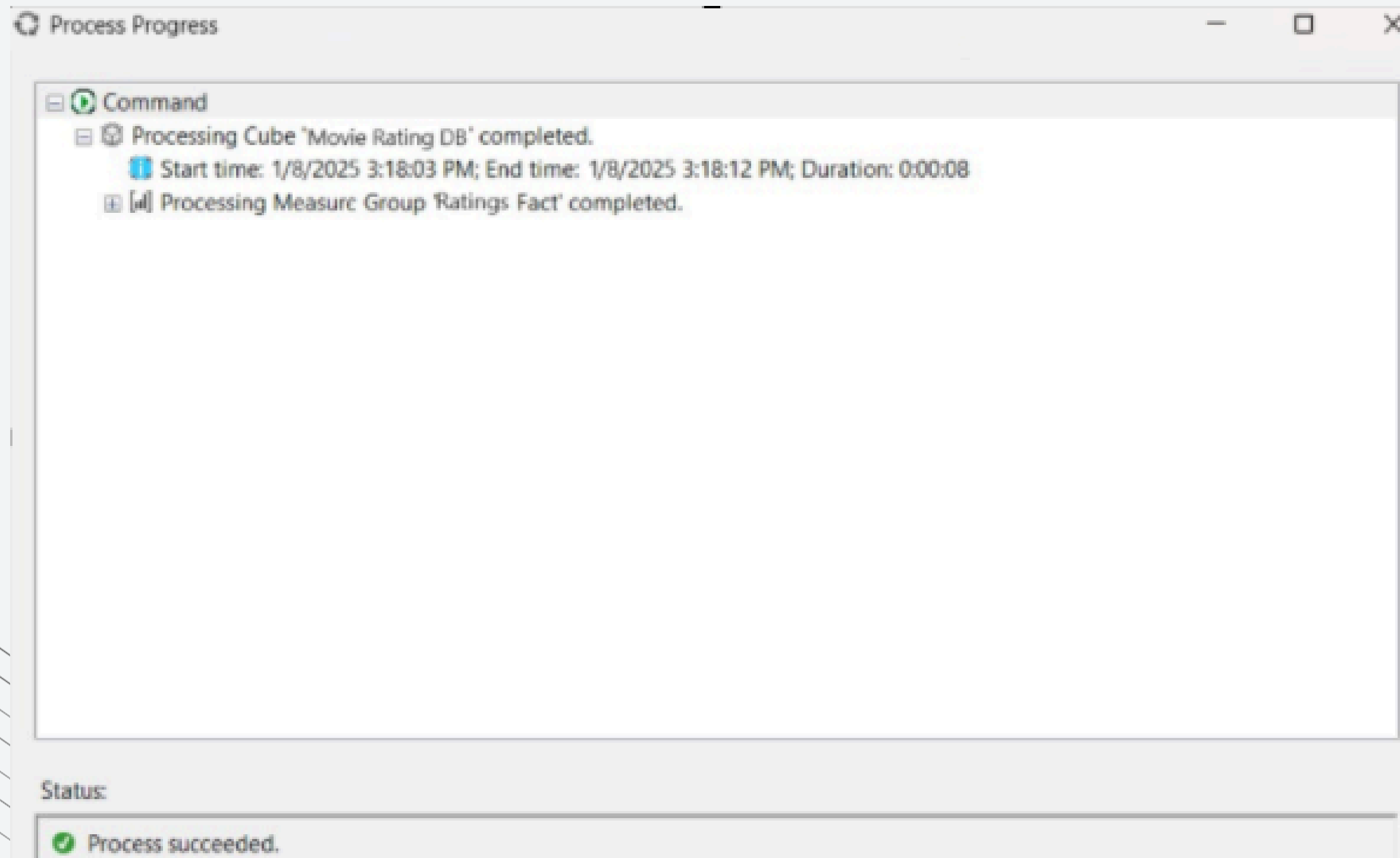
Final Data Cube schema



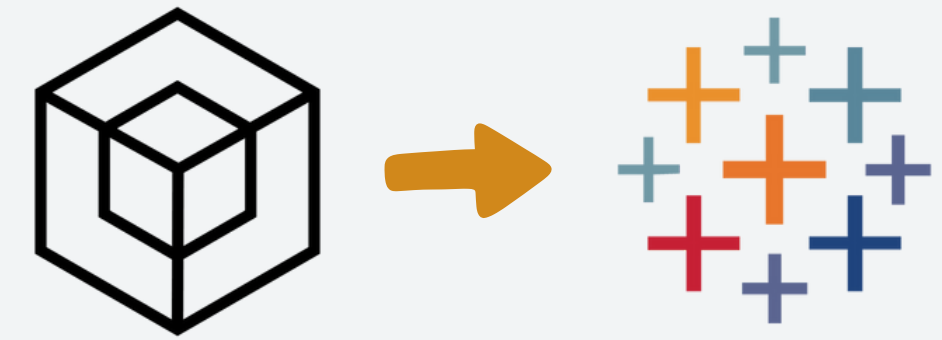
DATA CUBE CREATION

Successfully built and processed the cube.

Deployment completed without errors!

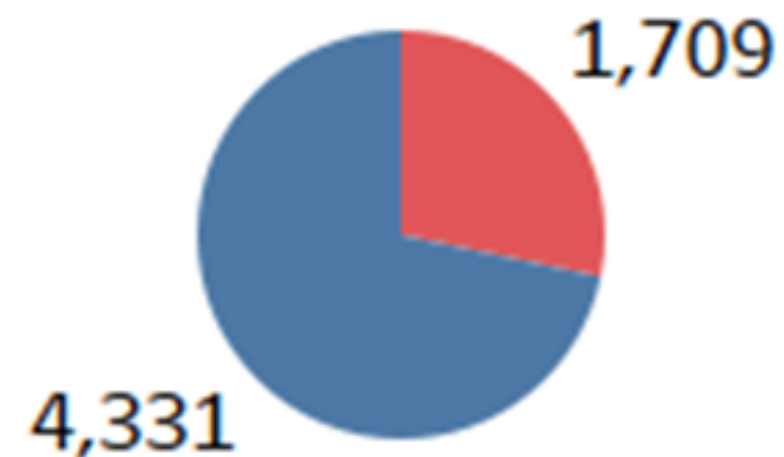


VISUALIZATION(1/11)

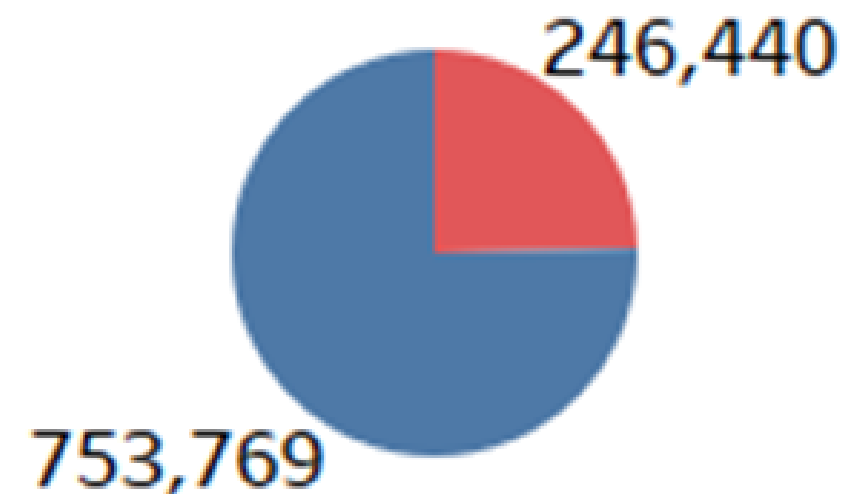


We connected the cube to Tableau

Users per gender



Ratings per gender

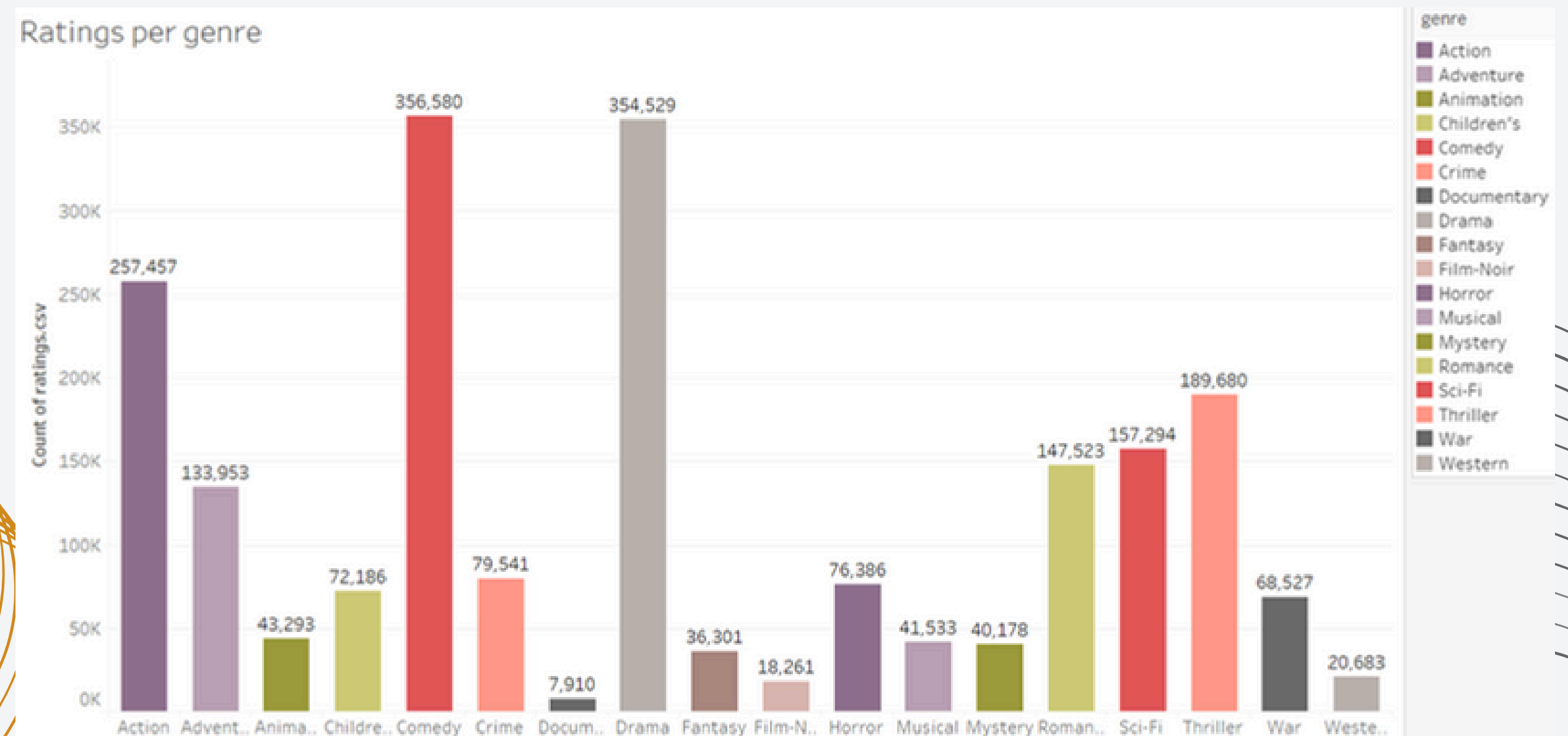


Only 25% of users are women.
Women contributed approximately 25% of all reviews.

VISUALIZATION(2/11)

Most reviewed genres:

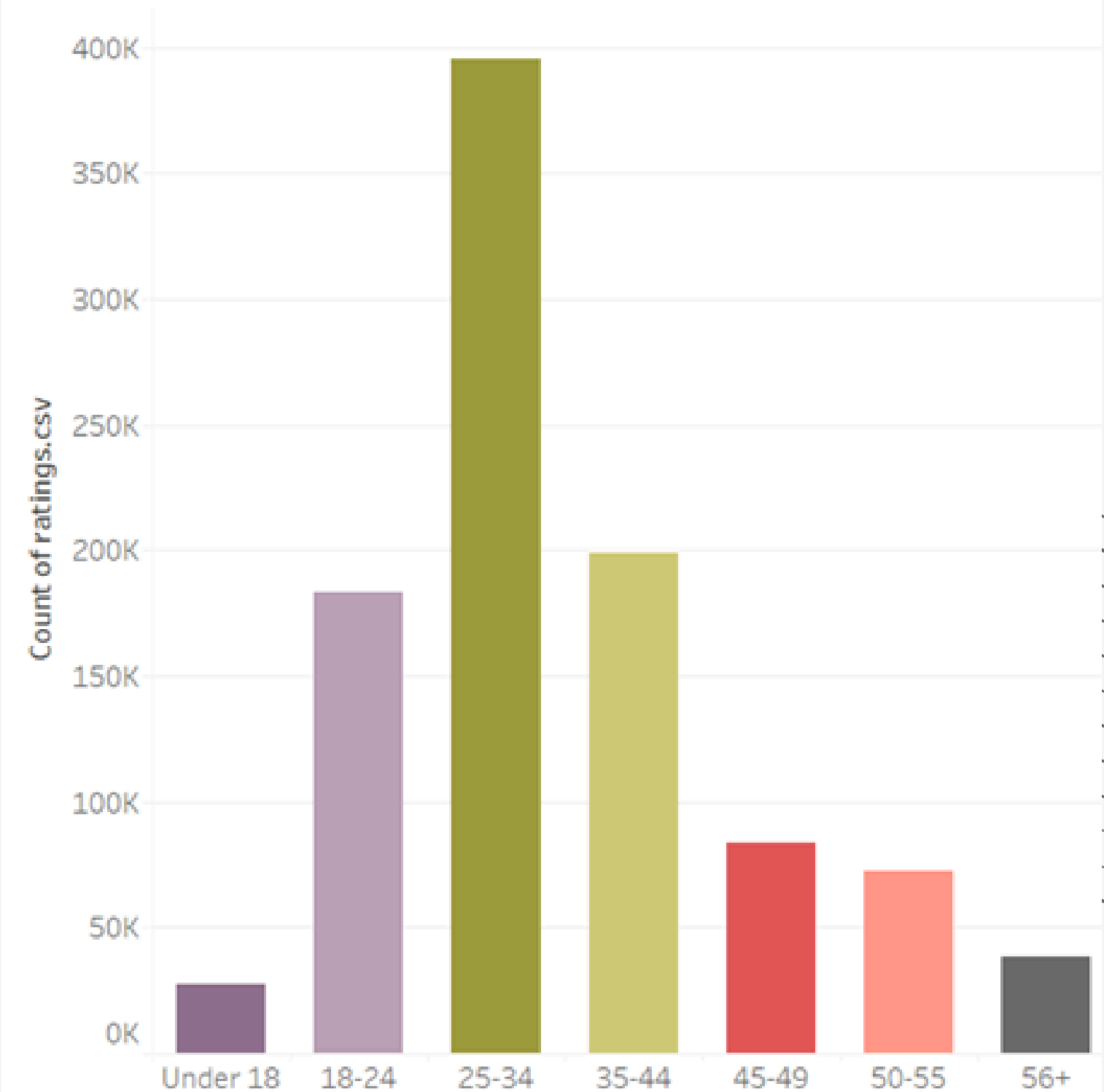
- Comedy
- Drama
- Action



VISUALIZATION(3/11)

25-34 age group contributed the most reviews, almost 40% (~400,000 reviews)

Ratings per age group and gender

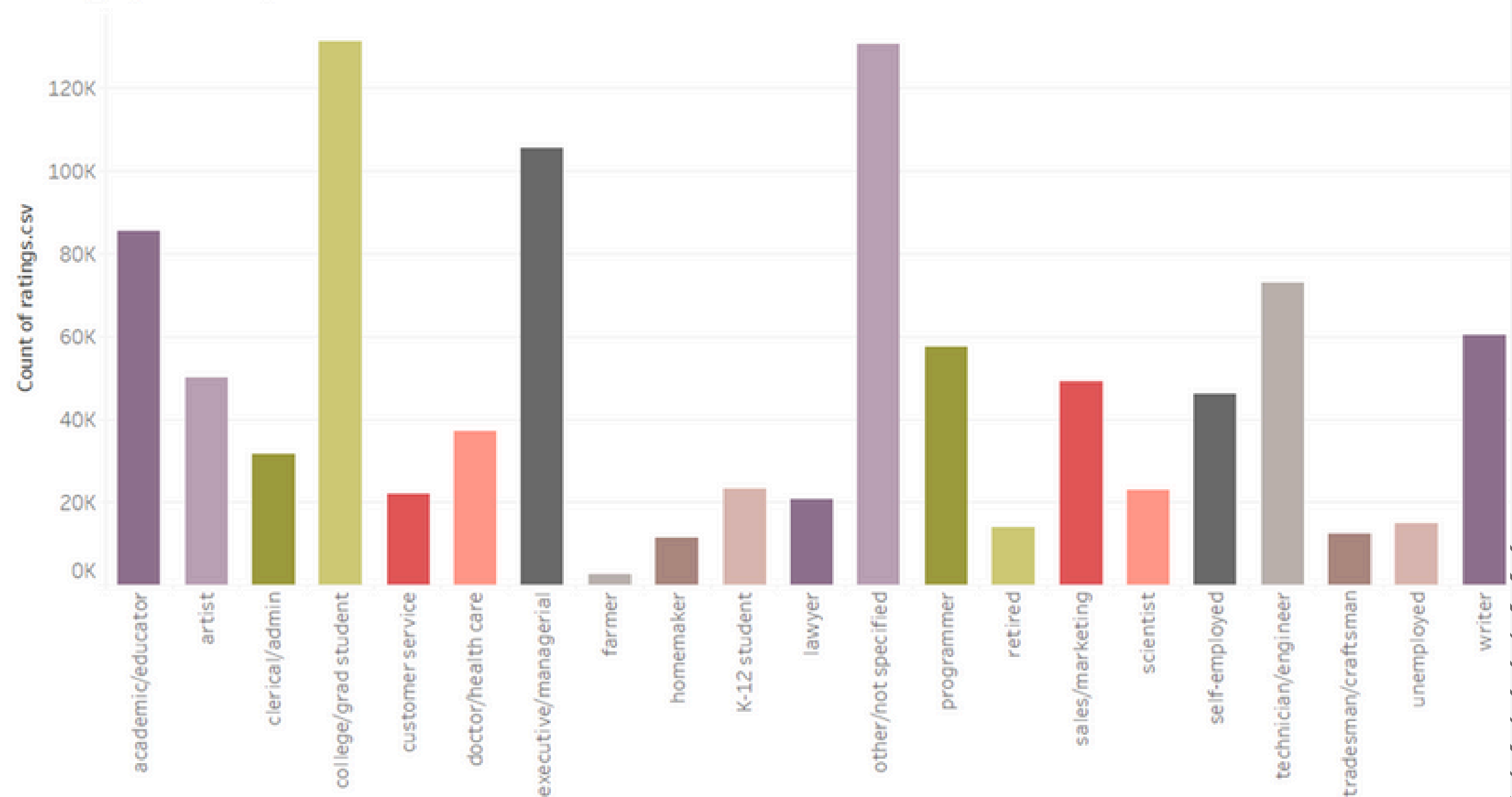


VISUALIZATION(4/11)

Top occupations for reviews:

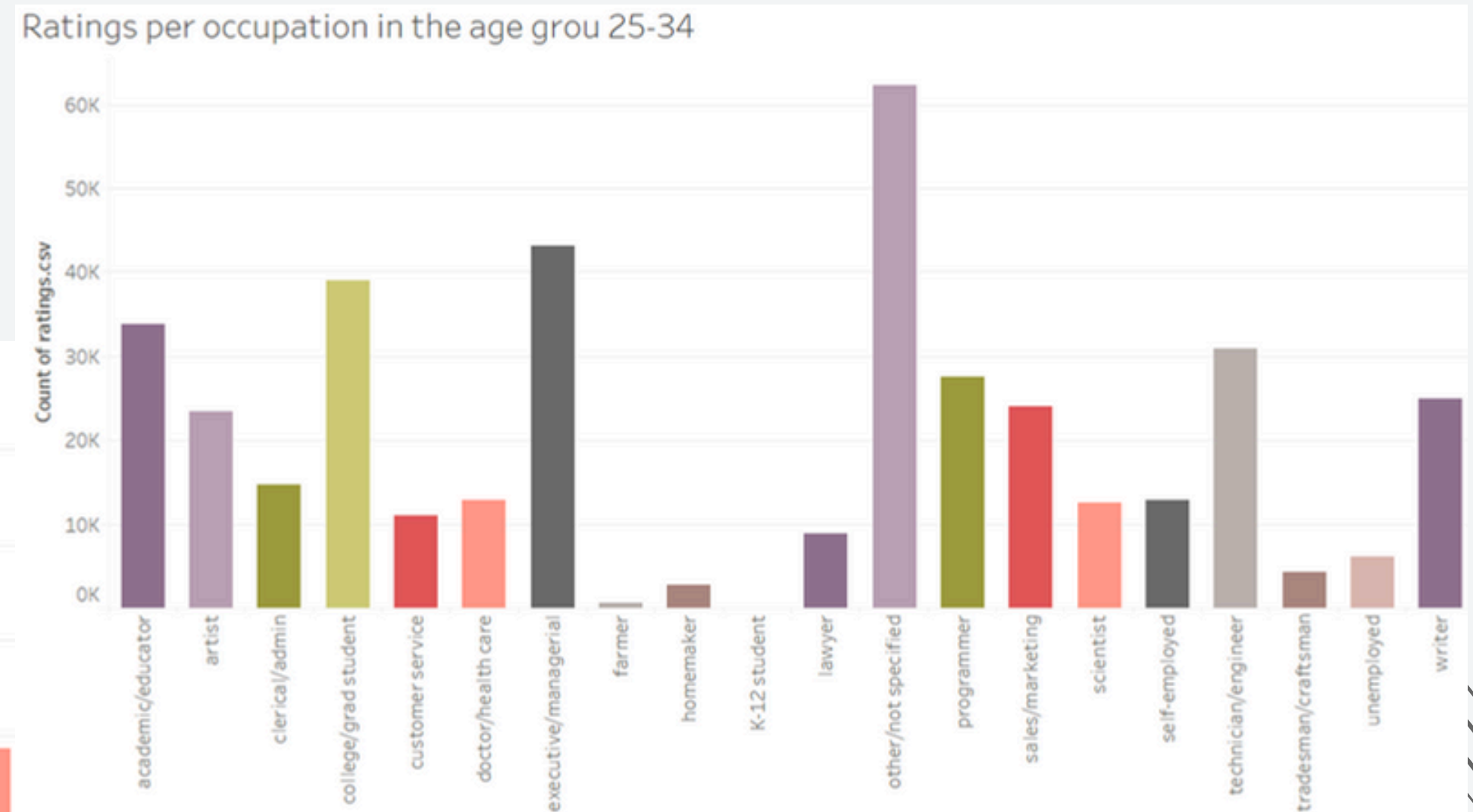
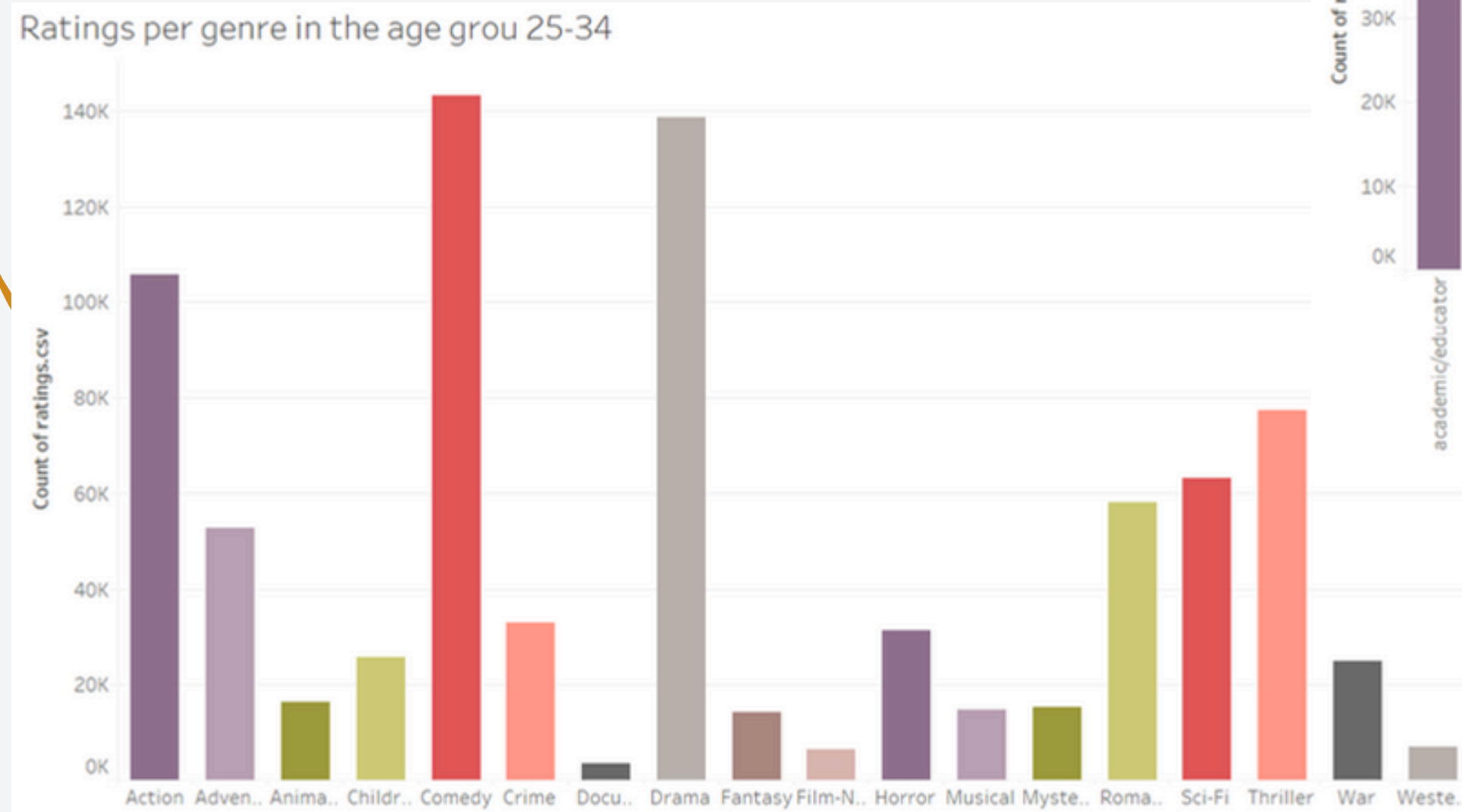
- College/Graduate Student
- Executive/Managerial
- Other/Not Specified

Ratings per occupation



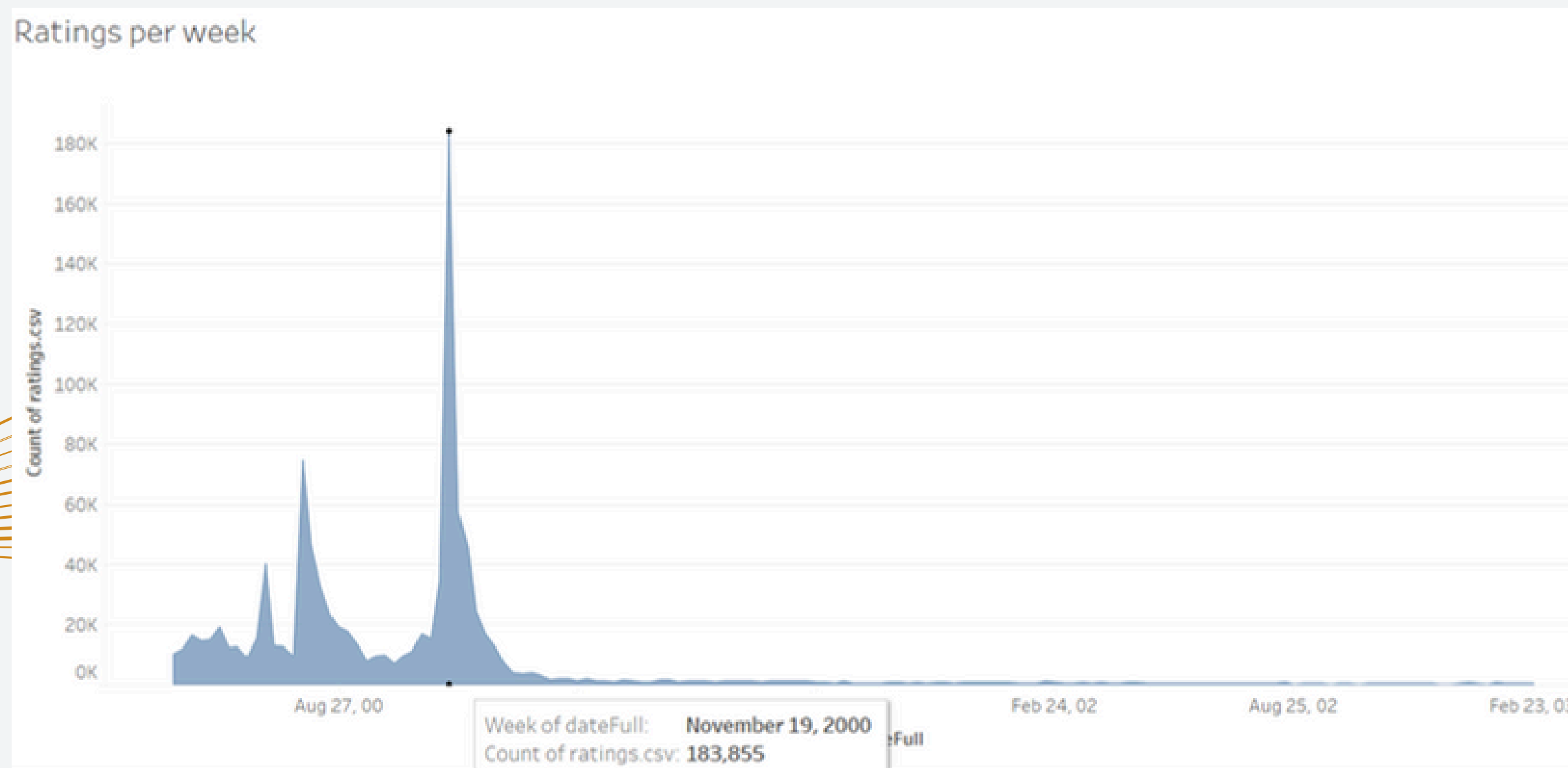
VISUALIZATION(5/11)

For the 25-34 age group:



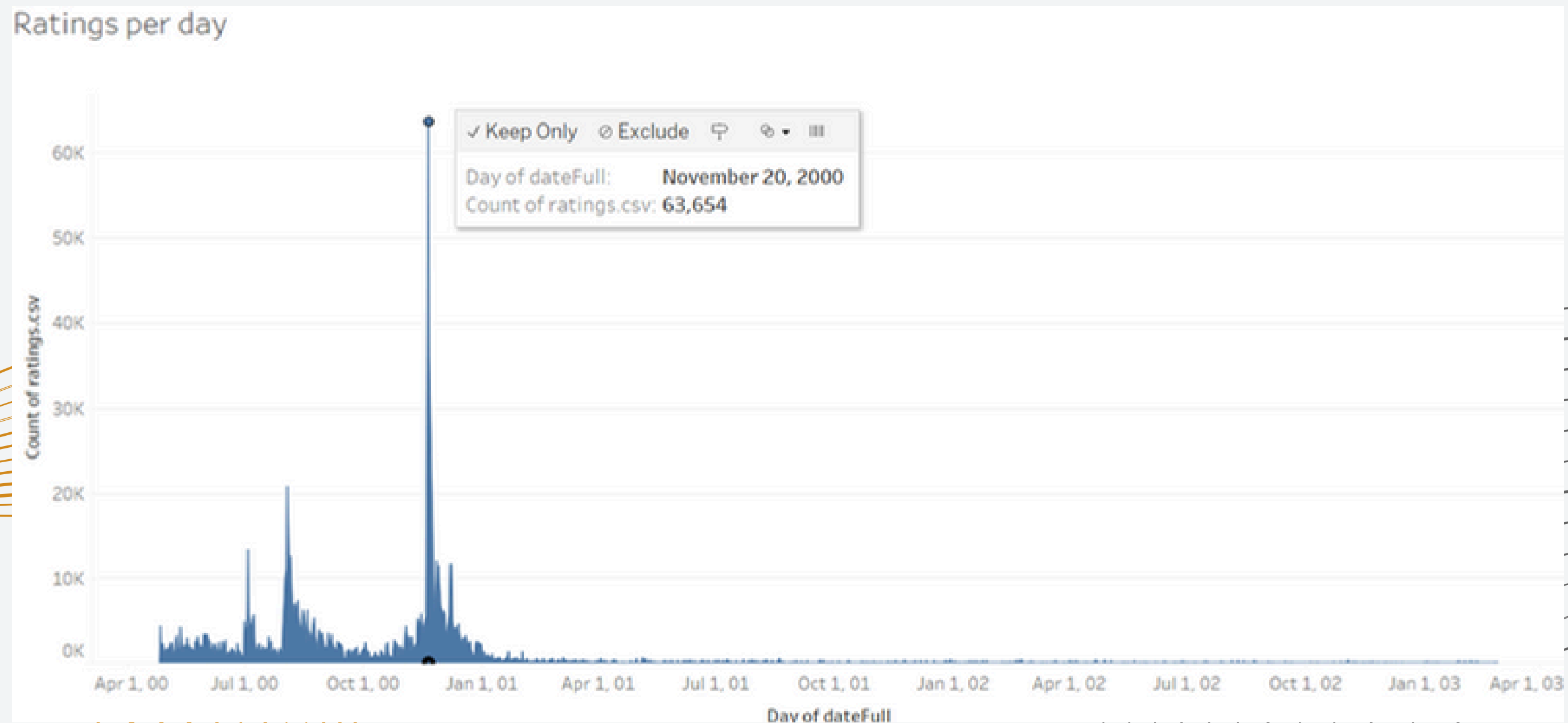
VISUALIZATION(6/11)

Peak week reviews: November 2000 (~183,000 reviews).



VISUALIZATION(7/11)

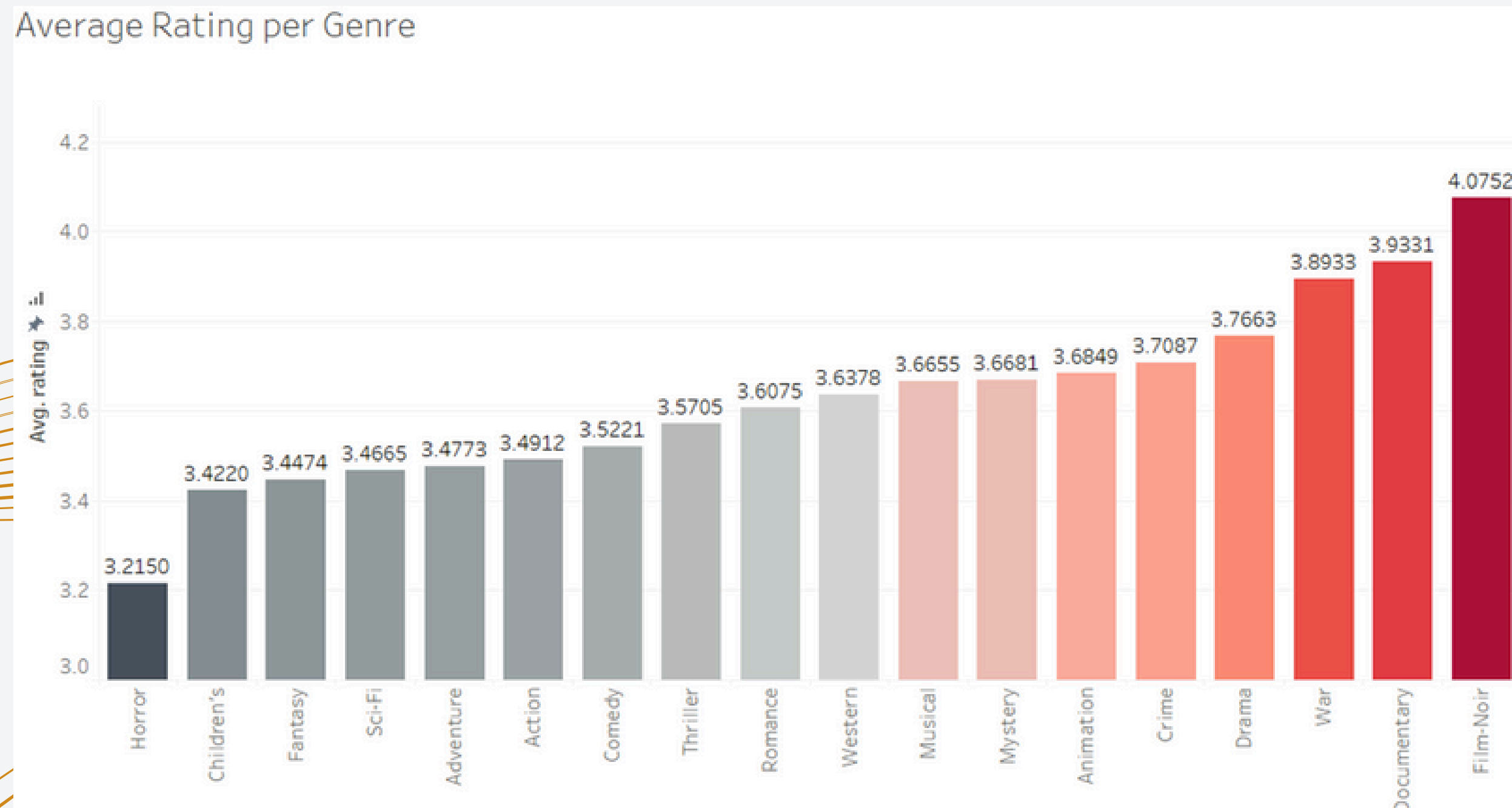
Most reviews on a single day: November 20, 2000 (~63,000).



VISUALIZATION(8/11)

Highest-rated genre: Film-Noir.

Lowest-rated genre: Horror.

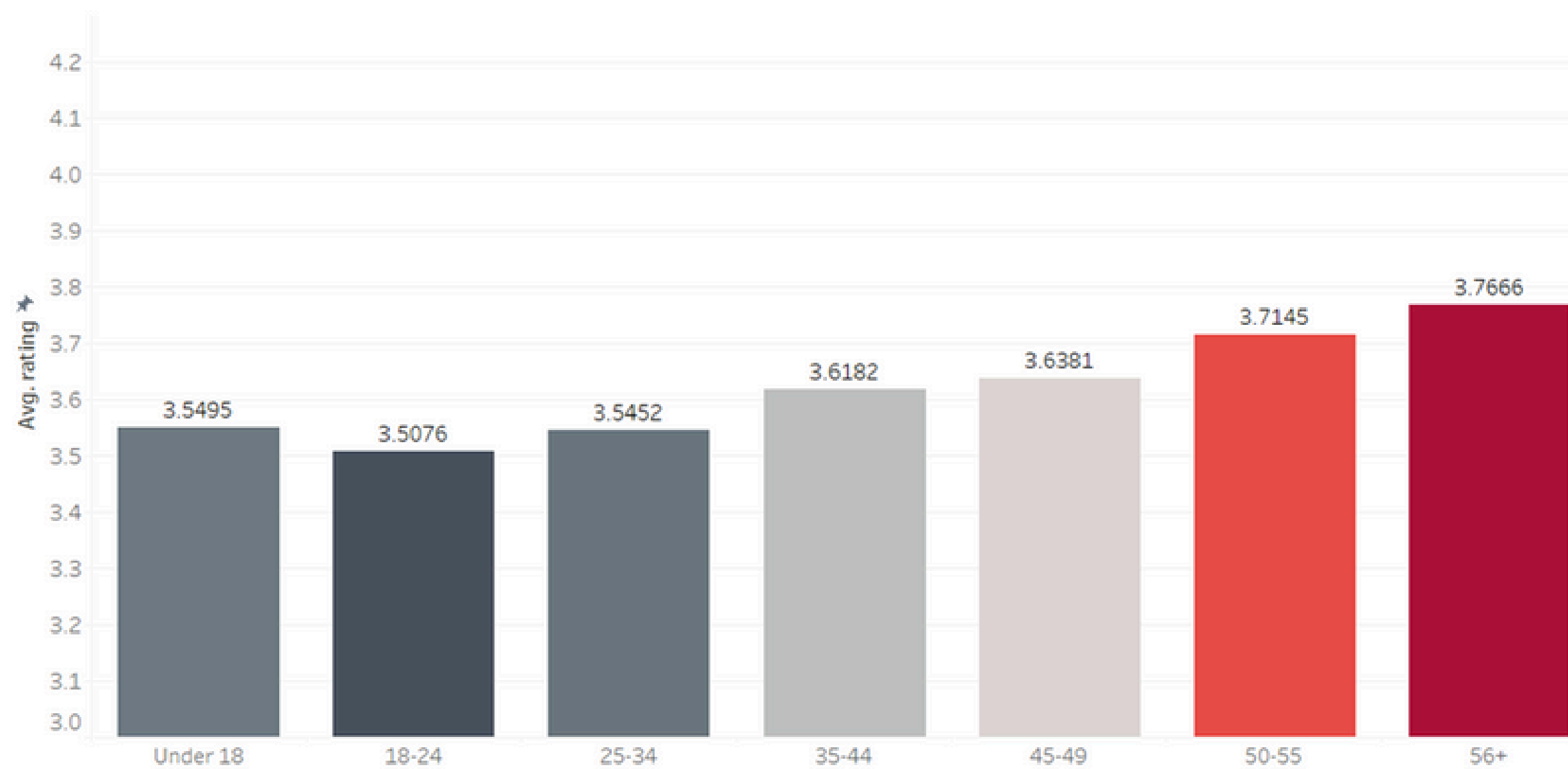


VISUALIZATION(9/11)

Strictest reviewers: Ages 18–24.

Most generous: Ages 56+.

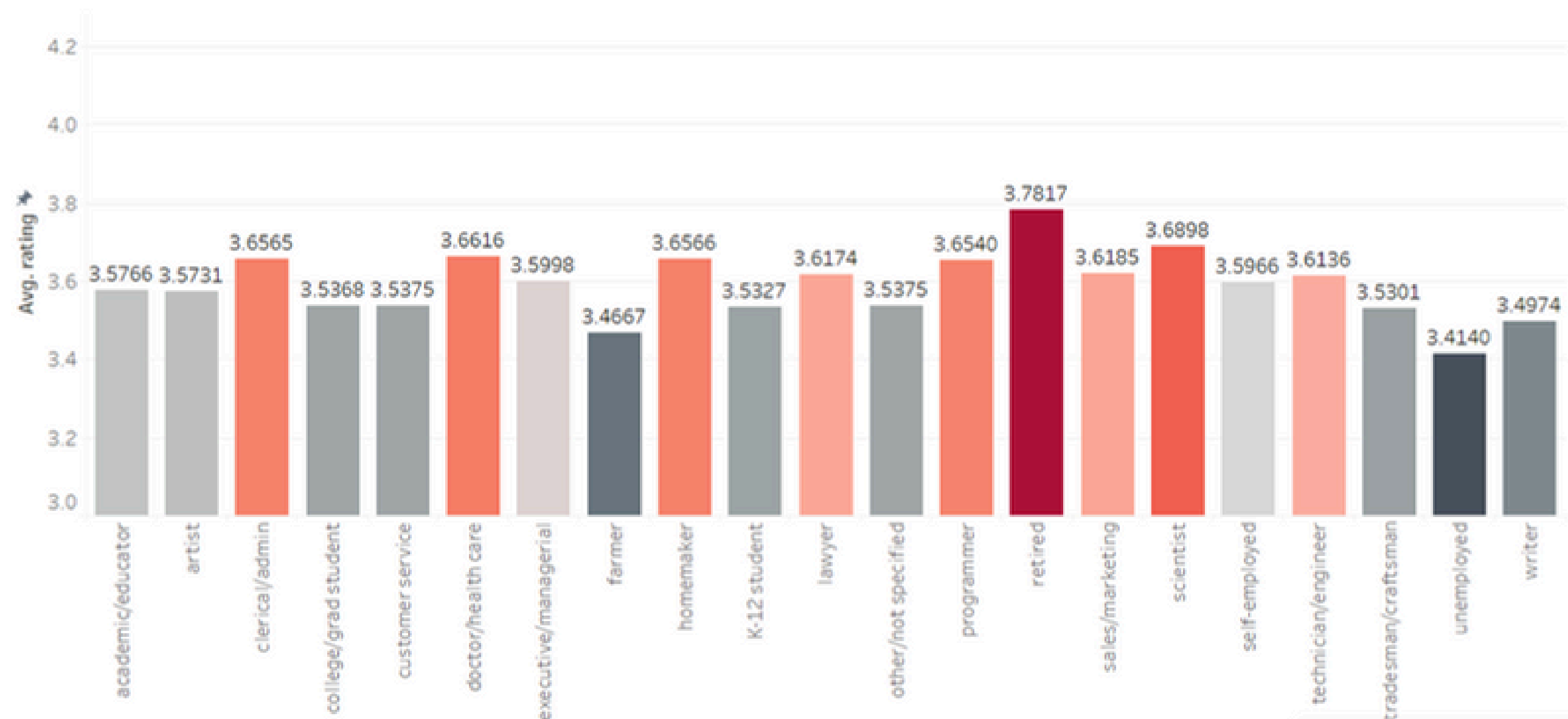
Average Rating per Age Group



VISUALIZATION(10/11)

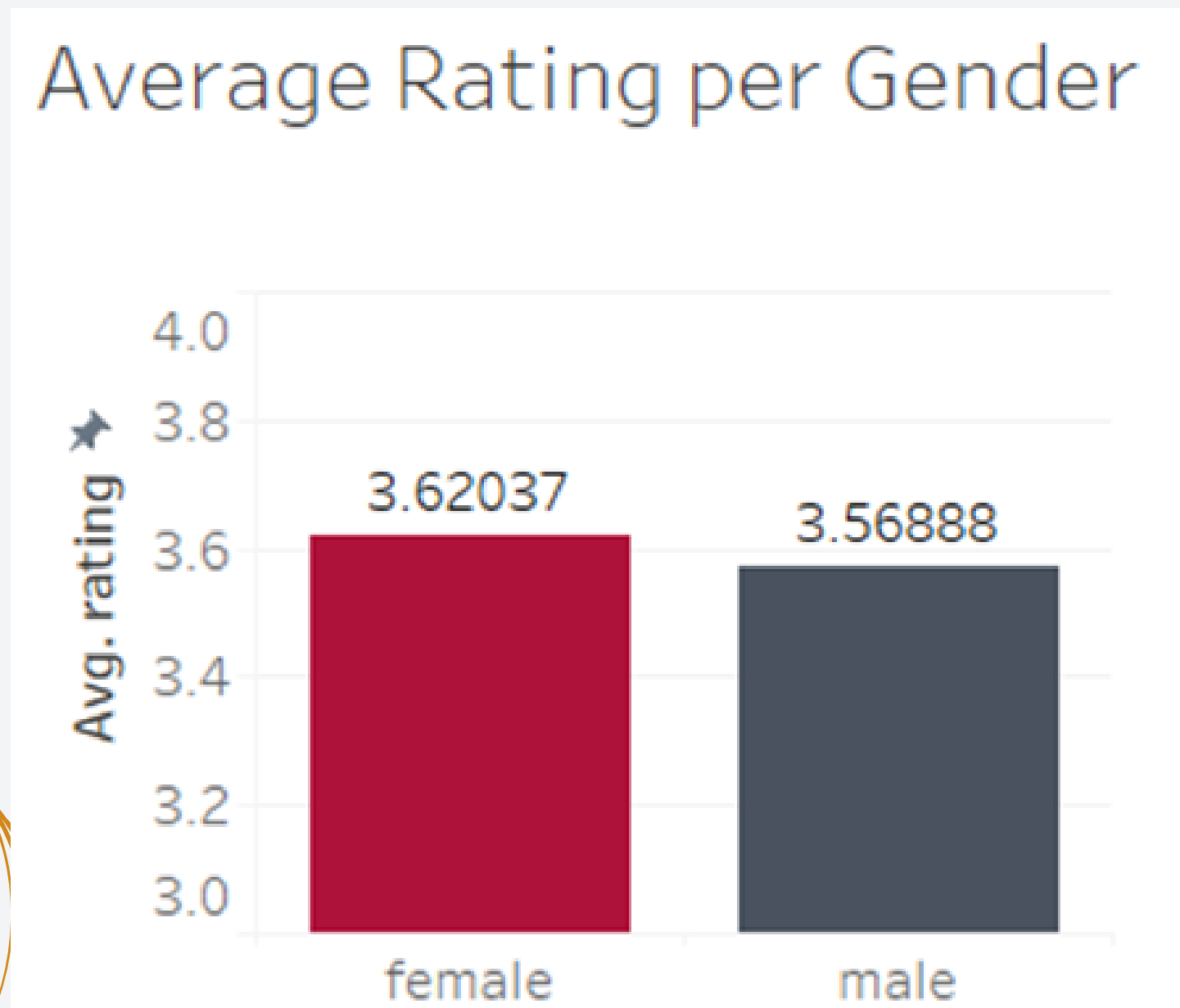
Strictest reviewers: Unemployed.
Most generous: Retirees .

Average Rating per Occupation



VISUALIZATION(11/11)

Women gave slightly better reviews than men.





Thank you!

Questions?

Alexandra Stathopoulou **8190156**

Stavros Siopis **8190320**